Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your answer. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A heat engine operates between **500 K** and **300 K** and absorbs **700 J**. **Find maximum ideal work** and **explain why actual work** is usually lower.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength **3 m** and frequency **2 Hz** travels at v=f λ=**6 m/s**. At the same speed, frequency **3 Hz** would require wavelength **2 m**.

Your turn: Draw both waves over the same **6 m** length and **label their frequencies**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



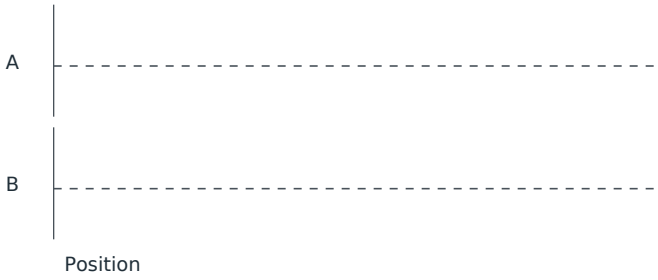
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two waves travel in the same ideal medium at the same speed. One has twice the frequency. **Compare wavelengths**.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

Wave speed

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A speaker produces a **425 Hz** tone in air where sound speed is **340 m/s**.
1. **Calculate its wavelength**.
2. **Find the frequency** that would produce a **0.50 m** wavelength in the same air.
3. **Sketch the relative crest spacings**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student writes v=f/λ. **Check units** and **explain why** that is not a speed.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An object at **360 K** sits in a vacuum facing surroundings at **240 K**. How can it cool, and are absolute-zero surroundings required?
Label temperatures, samples and relevant energy transfers.

Sentence starter: In my work, I used … to show … . I labeled it … .

Sample A

Sample B

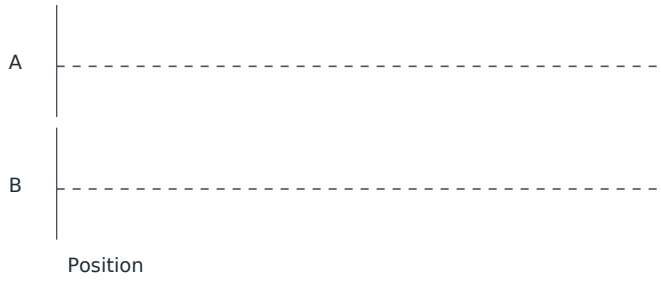
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

07 Exit check

A wave travels **12 m** in **0.40 s** and has frequency **50 Hz**. **Find speed** and wavelength **0**.

Sentence starter: I found … [unit]. This value means … in this situation.

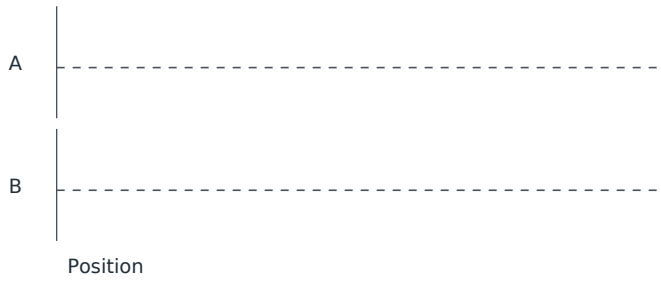


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A wave enters a medium where speed changes from **12** to **8 m/s**. Frequency stays **4 Hz**. **Find both wavelengths** and **explain why frequency** does not reset at the boundary.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③